

Research Article

Landscape management and planning as a spatial organization method connecting CES supply-demand assessment and sustainable tourism development

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ABSTRACT

Evaluating the balance between the supply and demand of Cultural Ecosystem Service (CES) in space is the key to linking ecosystem services to human well-being. However, due to its invisibility, it is difficult to quantify and is often ignored in ecosystem service evaluation. Based on social media photos, by integrating multisource data and using a clustering algorithm and the MaxEnt model, this paper maps and evaluates the supply-demand balance of three CES of aesthetic, historical and cultural, and leisure services in Suzhou, China. The result shows a wide-spread imbalance between supply and demand in Suzhou, among which the cultural heritage CES imbalance area as high as 48.1% accounts for the largest proportion, and the aesthetic imbalance area accounts for 38.4%, while the recreation imbalance area accounts for 37%. Building upon the assessment of supply and demand balance at the urban scale, we further identified the supply-demand imbalance at the scenic area scale by integrating local tourism development policies in Suzhou and selecting a typical scenic area, Xishan Island. Based on the urban and scenic area supply-demand assessment results, this paper proposes a multiscale CES supply-demand allocation strategy. This comprehensive approach not only furnishes valuable guidance for local decision-making and management within the realm of sustainable tourism but also holds potential for wider applicability to diverse urban areas worldwide that share analogous contextual backgrounds.

Management implications: CES supply and demand assessment can help identify the CES supply-demand imbalance areas of tourist destinations. Landscape management and planning is a spatial organization method to balance CES supply-demand mismatches in tourist destinations. Optimizing landscape structure at multiple scales will be of instructional significance for the protection and development of tourist destinations. A 'CES assessment and mapping – landscape management and planning – tourism protection and development' system is beneficial to achieve the sustainable development of tourism.

1. Introduction

Tourism connects people with the natural environment and has a positive effect on people's mental health and quality of life. CES illustrates the benefits and comforts that nature provides to people (MEA, 2005). A deeper understanding of CES would clarify the relationship of tourism between nature and human well-being, providing value for tourism destination protection and development (Smith & Ram, 2017; Willis, 2015). At the same time, the trade-off between tourism protection and development is a challenge to CES assessment and mapping (Bachi, Ribeiro, Hermes, & Saadi, 2020; Hermes & Saadi, 2020; Mouttaki et al., 2021; Ribeiro & Ribeiro, 2016). Therefore, analyzing the

supply and demand situation of CES in space plays an important role in balancing protection and development strategies (Castro et al., 2014; Meng et al., 2020). Landscape is considered a medium to study the relationship between CES and tourism and is regarded as a fundamental element of tourism (Smith & Ram, 2017). Landscape management and planning, on the other hand, provide guidance on the spatial optimization of tourism (Chen, Thapa, Kim, & Yi, 2017). To balance CES supply-demand mismatches, there is an urgent need for a complete set of CES assessments and mapping - landscape management and planning - tourism protection and development systems to achieve sustainable tourism through protected development (Fig. 1).

In recent years, the relationship between tourism and Cultural

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Ecosystem Service (CES) has garnered significant attention from researchers worldwide (Ram & Simth, 2019). This intersection has been explored through a variety of methodologies, including both monetary and non-monetary approaches, to quantify the value of CES in diverse tourist destinations (Bastarz & Biondi, 2011; McCain & Ray, 2003; Smith & Ram, 2017; Thapa et al., 2020). Despite the progress made in understanding this connection, earlier studies often fell short in providing fine-scale spatial insights into CES distribution, which is crucial for informed decision-making in tourism management. A notable development in addressing this gap is the advent of CES mapping, a method for visualizing the spatial distribution of CES. Researchers have leveraged technologies such as participatory mapping and geographic information systems, notably ArcGIS, to chart tourists' perceptions of CES regions (Burkhard, Kandziora, Hou, & Müller, 2014; Swetnam et al., 2011; Zoderer, Tasser, Erb, Stanghellini, & Tappeiner, 2016).

These mapping techniques have found applications in a variety of settings, including natural parks (Ribeiro & Ribeiro, 2016), agricultural landscapes (Van Berkel & Verburg, 2014), coastal regions (Ghermandi, Camacho-Valdez, & Trejo-Espinosa, 2020; Ruskule, Klepers, & Veide-mane, 2018), mountainous terrains (Zoderer et al., 2016), and urban tourist hubs (Nahuelhual, Carmona, Lozada, Jaramillo, & Aguayo, 2013). While these methods have demonstrated their value, they often require considerable time and resources, limiting their scalability (Ghermandi et al., 2020). However, the rise of social media has introduced a new data source for CES research (Da Mota & Pickering, 2020; Figueroa-Alfaro & Tang, 2017). Artificial recognition (Clemente et al., 2019), deep learning (Vieira et al., 2021; Cardoso et al., 2022; Egarter Vigl et al., 2021), clustering algorithms (Fox, Graham, Eigenbrod, Bullock, & Parks, 2021; Richards & Tunçer, 2018), and other data analysis approaches have been harnessed to achieve several key objectives, such as assessing spatial CES supply, investigating the influence of environmental and social factors on CES distribution, and creating spatial models of tourist preferences for CES (Mouttaki et al., 2021; Richards & Friess, 2015; Thiagarajah, Wong, Richards, & Friess, 2015; Zhao, Li, Liu, Cuan, & Zhang, 2019).

Recent research trends in the field are primarily centered on the precise spatial mapping of CES supply and demand. The aim is to provide decision-makers and planners with more granular and location-specific information to better manage, plan, and develop tourist destinations (Kalinauskas et al., 2023; Zhao, You, Lin, & He, 2023). This study defines the supply and demand of CES as follows: the supply of CES is defined as the potential provided by the ecosystem to deliver cultural and entertainment services, while the demand for CES is defined as the perceived need for such services by individuals influenced by socio-economic factors (Bing et al., 2021; Burkhard et al., 2012; Casado-Arzuaga, Onaindia, Madariaga, & Verburg, 2014). Understanding

these supply-demand dynamics and the associated spatial heterogeneity and imbalances is instrumental in making tourism more sustainable and efficient (Crossman et al., 2013; Wu, Zhao, Shen, Qin, & Liu, 2018).

Furthermore, addressing imbalances in CES supply and demand is a priority. Identifying areas where there is either an excess supply or demand for CES can guide interventions aimed at achieving more sustainable utilization of these services (Lorilla, Kalogirou, Poirazidis, & Kefalas, 2019; Meng et al., 2020; Shi, Chen, Liu, Geng, & Zhang, 2022). Finding new tourism value in imbalanced areas and shifting tourism pressure to other areas has proven to be beneficial for promoting the balance of CES supply-demand mismatches (Clemente et al., 2019; Meng et al., 2020; Ruskule et al., 2018; Baró, Gómez-Baggethun, & Haase, 2017). However, despite these advancements, the inherent intangibility and subjectivity of CES continue to present challenges, particularly in spatial planning for tourist destinations (Clemente et al., 2019; Richards & Friess, 2015), where a more holistic approach is needed to ensure the efficient deployment of tourism resources.

As CES research has progressed, scholars have recognized the significance of integrating CES into landscape management and planning systems for the tourism industry (Castro et al., 2014; González-García, Palomo, González, López, & Montes, 2020; Xu et al., 2022). They have explored this integration on three distinct levels: (1) Incorporating CES assessment into the landscape management and planning framework (Albert et al., 2016; Bezák, Mederly, Izakovičová, Špulerová, & Schleyer, 2017); (2) Utilizing landscape indicators to enhance the sustainability of tourist destinations (De Vreese, Leys, Fontaine, & Dendoncker, 2016; Frank, Fürst, Koschke, & Makeschin, 2012); and (3) Preliminary discussions on integrating CES assessments through specific mapping exercises (Schirpke, Timmermann, Tappeiner, & Tasser, 2016). These efforts are vital for optimizing the coexistence of cultural and natural elements in landscapes and ensuring that tourism development aligns with broader sustainability goals.

The intricate relationship between the spatial equilibrium of CES and landscape characteristics, as well as management practices, underscores the crucial role of effective landscape spatial management and planning in fostering sustainable tourism development. Accordingly, this research endeavors to address several pivotal inquiries:

- How can an assessment methodology leveraging social media photo data discern and identify diverse imbalanced regions characterized by CES supply-demand disparities across multiple spatial scales?
- What analytical approaches can be employed to meticulously analyze the spatial distribution patterns of oversupply and under-supply within CES imbalanced regions, elucidating the underlying causal factors driving such dynamics?

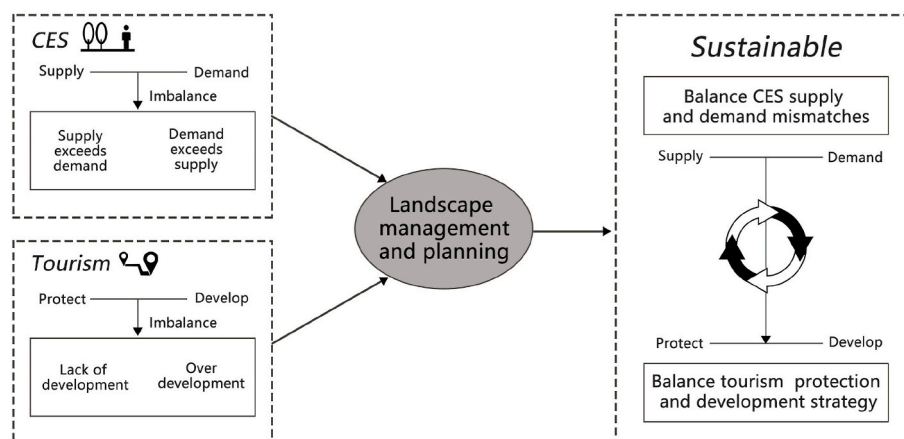


Fig. 1. The landscape management and planning play an important role in connecting CES and tourism. The balance of tourism protection and development strategies through balancing CES supply-demand mismatches to achieve sustainability.

- What specific landscape management and planning strategies are indispensable for addressing CES supply-demand imbalances at the regional scale of Suzhou, considering the nuanced contextual factors at play?
- How can the evaluation and analysis of CES supply-demand balance at the scenic spot scale of Suzhou provide actionable insights to optimize critical landscape features and spatial elements, enhancing the overall visitor experience and sustainability of the destination?
- How can the integration of CES spatial assessment with landscape spatial organization strategies culminate in a comprehensive framework, enabling effective multi-scale supply-demand allocation and facilitating the realization of sustainable tourism development goals?

Although the employment of landscape management and planning as a method of supply-demand spatial allocation and spatial organization helps to achieve a dynamic balance in the spatial distribution of CES supply-demand in tourist destinations (Plieninger et al., 2015), in the process of incorporating CES into the landscape management and planning system, there are still practical challenges, such as insufficient connectivity between CES assessment and landscape characteristics and incompatibility between CES assessment and site scale (Cheng, Van Damme, & Uytendhove, 2021). Therefore, further incorporation of CES into landscape management and planning systems needs to focus on different scales, conduct detailed analysis of the landscape elements in and around the site (Richards & Friess, 2015), and propose corresponding landscape management and planning strategies to enhance the attractiveness of tourist destinations or carry out monitoring and protection. Multiscale assessments at both the regional and scenic spot scales can help to achieve differentiated management within regions (Zhou et al., 2022) and formulate practical strategies for the protection and development of tourist destinations for decision makers.

Taking Suzhou as an example, based on social media data and equipped with MaxEnt and GIS, we carried out multiscale supply and demand mapping of three types of CES: aesthetic, cultural heritage, and recreation services. This study aims to develop a comprehensive system of 'CES assessment and mapping - landscape management and planning - tourism protection and development'. The establishment of the system includes four purposes: (1) using landscape management and planning as a medium of spatial organization to establish the connection between CES supply-demand assessment and the sustainable development of tourism destinations; (2) through the optimization of landscape spatial structure, the imbalanced area of supply-demand of CES in Suzhou is deployed to promote the coordinated development of tourism areas to the overall integration process of tourism in Suzhou; (3) through assessment and mapping of the two different scales of city and scenic spots to promote mutual cooperation between governments at all levels; and (4) realizing the differentiated management of tourism areas through the balance between protection and development strategies. The implementation of this system includes the following three stages: (1) CES supply and demand assessment and mapping and identification of the CES supply-demand imbalance area; (2) optimization of the landscape spatial structure through landscape management and planning methods and allocation of the CES supply-demand imbalance area; and (3) formulation of appropriate tourism protection and development strategies according to the multicore and networked landscape spatial structure layout to realize the sustainable development of tourism destinations.

In summary, this research aims to contribute to the existing theoretical framework of spatial assessment for CES and sustainable tourism development by identifying CES supply-demand imbalances at different spatial scales and analyzing the underlying mechanisms driving these imbalances in tourist destinations. Furthermore, this study innovatively introduces differentiated landscape management and planning strategies to effectively address the identified imbalances in CES. Ultimately, the landscape spatial organization strategies serve as a pivotal medium

bridging CES supply-demand assessment and sustainable tourism development, aiming to formulate a comprehensive framework applicable to diverse tourist destinations and foster sustainable development practices on a global scale.

2. Study area

Suzhou city is located in the middle of the Yangtze River Delta, between 119°55'E ~ 121°20'E and 30°47'N ~ 32°02'N, an area with low and flat land, where the water networks are dense, and they have become unique ecological and cultural resources. The CES of aesthetic, cultural heritage, and recreation has become the core demand of the public (Yushu, Zhenyu, & Zhuoran, 2021), and it is also the key to shaping spatial vitality in the urban landscape of Suzhou. The advancement of the "regional integration" initiative provides opportunities for the co-ordinated development and optimization of the CES supply and demand balance. This paper selects the township-level administrative units in Suzhou as the basic research unit, which includes a total of 128 townships (Fig. 2).

3. Materials and methods

3.1. Research framework

Employed with geotagged photos obtained from social media, we combined environmental data to map the CES supply of the three types using MaxEnt. In addition, we also use the ArcGIS kernel density tool to map the CES demand. By comparing the CES supply map and the CES demand map, the CES supply-demand balance map is calculated, and the CES supply-demand imbalance areas are extracted. Through the identification of landscape elements in the two different types of CES over-supply and over-demand areas, we developed corresponding demand-oriented and supply-oriented landscape management and planning strategies to optimize the landscape structure. By formulating landscape management and planning strategies, the balance between tourism protection and development could be promoted; thus, sustainable development for tourism destinations could be achieved (Fig. 3).

3.2. Data collection and processing

3.2.1. Photo dataset

Although social media photos from Flickr have been widely used in CES mapping (Schirpke, Tasser, Ebner, & Tappeiner, 2021; Sinclair, Ghermandi, Moses, & Joseph, 2019; Yoshimura & Hiura, 2017), Flickr users in China are limited. "Six Foot" serves as an open platform tailored for travelers, combining the functions of recording, browsing, and sharing outdoor travel routes. Currently, it has managed to engage outdoor enthusiasts across various age groups, including the elderly, middle-aged, and youth, hailing from both local and non-local regions. In contrast to other platforms, "Six Foot" distinguishes itself by offering users the capability to spontaneously capture geotagged photos. These photos serve as a reflection of travelers' preferences for various CES, and the geographical coordinates associated with these photos provide valuable insights into the spatial distribution of CES preferences. We therefore selected the local outdoor activity platform 'six-foot' as our geotagged photo source, then screened the data from the platform. The filter criteria were "Suzhou", "Walking", upload time between "2021.7 to 2022.7", and "Itinerary with photos". We then write scripts in Python and log in to the 'six-foot' platform (<http://www.foooot.com/>) to crawl 104,207 geotagged photos. Random sampling was performed on the crawled photos, and a sample with 5000 data volumes was gathered. To ensure the accuracy of the results, photos unrelated to the CES supply were manually removed, including photos with people as the subject, such as selfies; items unrelated to landscapes, such as signs; display boards, etc.; unrecognizable and poor-quality photos; photos taken for special purposes, such as river water quality surveys, safety equipment

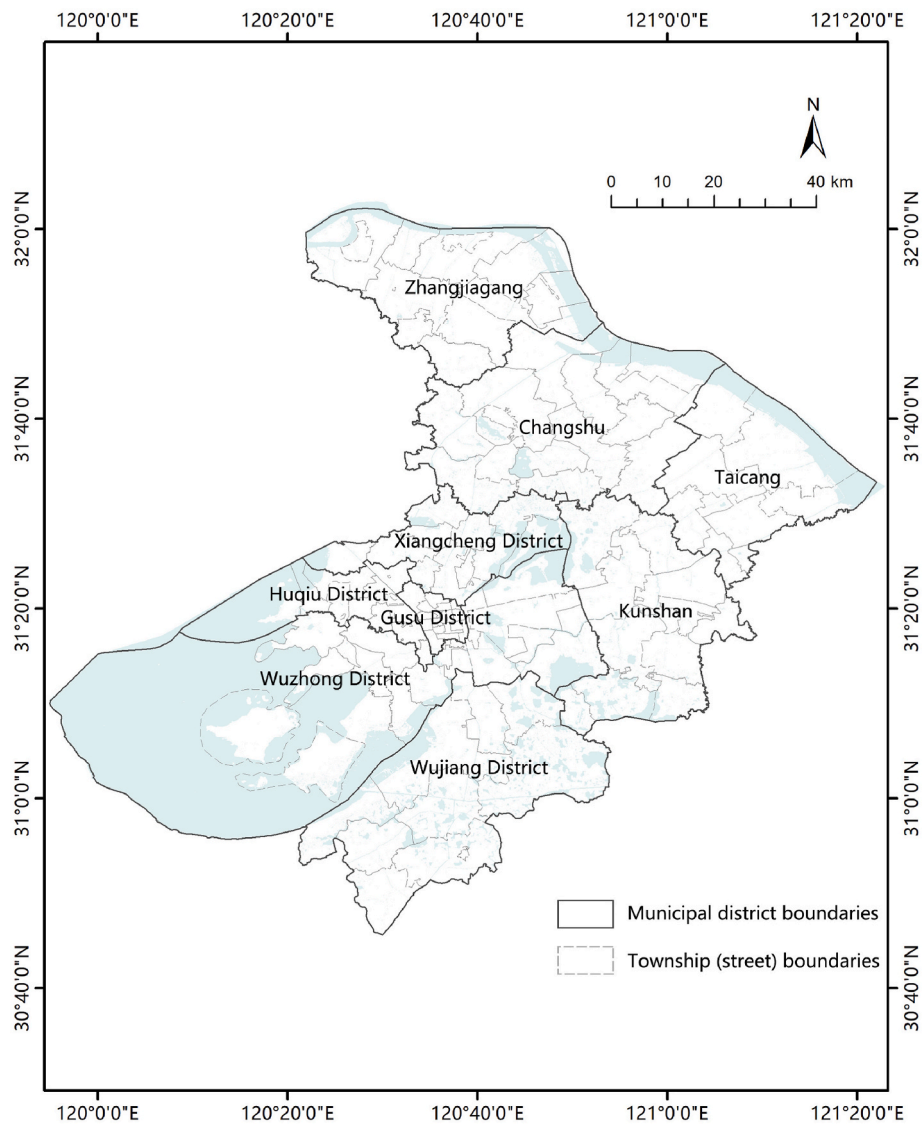


Fig. 2. Location and administrative map of the study areas. In this study, the administrative (or township) unit of Suzhou was used as the mapping unit of CES supply and demand.

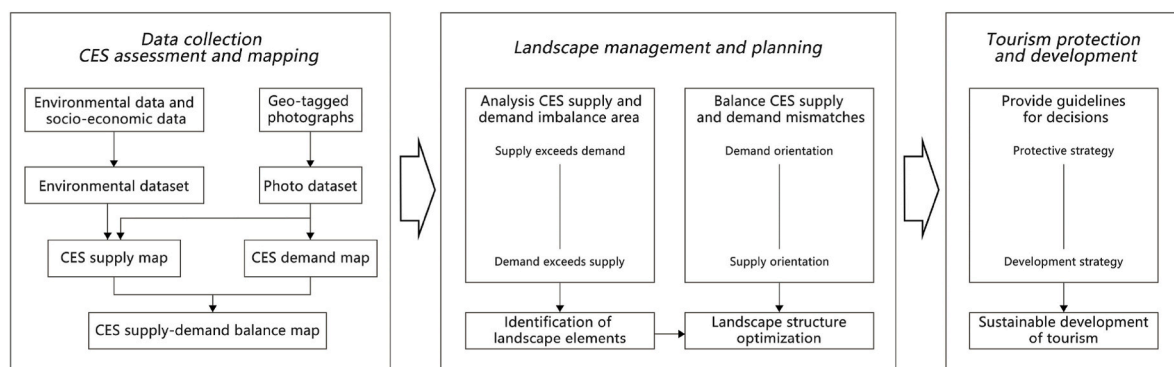


Fig. 3. Illustrating research design aims and expected results.

inspections, road signs, etc. (Tenerelli, Demšar, & Luque, 2016), and each user retains no more than 2 photos. Finally, 279 available photo data points were obtained. According to the classification and definition in the Millennium Ecosystem Assessment, three main types of CES are extracted for our analysis (MEA, 2005). We then divided the photos into

three types (aesthetic, cultural heritage, and recreation) accordingly and at the same time allowed the same photos to belong to multiple categories (Table 1). After classification, 159 photos of aesthetic, 55 photos of cultural heritage, and 65 photos of recreation were obtained (Fig. 4).

Table 1
Classification of CES.

CES Type	Definition	Example	Photo
Aesthetic	The aesthetic enjoyment that people experienced (from different aspects of the ecosystem, such as scenery, sights, sounds, etc.)	The photos showing natural and culture landscapes of the ecosystem	
Cultural heritage	Significant historical landscapes (natural and culture) maintained by the ecosystem or historical remains with significant cultural value	The photos showing tangible (buildings, cultural landscapes, species associated with local culture) or intangible cultural heritage (traditional customs, folklore)	
Recreation	Recreational services provided by ecosystems based on natural or seminatural landscapes	The photos of recreational activities (hiking, mountain-climbing)	

3.2.2. Environmental dataset

The environmental data and socioeconomic data that could affect the supply of the three types of CES are screened out by integrating elements such as nature, society, and human activities, and an environmental dataset is established, including 1) distance to the CES sites; 2) elevation; 3) distance to roads; 4) distance to rivers and lakes; 5) vegetation coverage index; 6) landscape diversity index; 7) habitat quality index; 8) land use type; 9) population density; and 10) night-time satellite data. All data are from open source databases (Table 2), and all variables are continuous. A unified coordinate system and pixel size are set using ArcGIS, and the asc. format is output for MaxEnt recognition.

3.3. CES assessment and mapping

3.3.1. Mapping CES supply

This study used MaxEnt 3.3 to assess and map the potential supply of CES (Arslan & Örüçü, 2021; Clemente et al., 2019). The software used machine learning to estimate the probability distribution by finding the probability distribution of maximum entropy for a given observation point. The observed points were called existence data. In this study, the photo dataset obtained from the 'six-foot' was used as the existing data,

the three types of photo points and environmental data after screening were input into MaxEnt for calculation, and the number of runs was set to 10. Average AUC of the run generated as the final model to visualize potential supply for the three types of CES.

3.3.2. Mapping CES demand

The selected geotagged photo dataset could reflect people's perception of CES spatially (Ros-Candeira, Moreno-Llorca, Alcaraz-Segura, Bonet-García, & Vaz, 2020; Richards & Tunçer, 2018). The kernel density analysis method is widely used in hotspot prediction and can convert discrete points in space into raster density maps (Wang et al., 2021). Subsequently, the selected photo spots of the three types were imported into ArcGIS as a dataset for kernel density analysis. The visitors' demand based on the 'six-foot' platform was evaluated, and the existing CES demand distribution map was established, which would reflect the tourist attraction level in Suzhou from July 2021 to July 2022.

3.3.3. Mapping CES supply-demand balance

Taking the township-level administrative unit as the basic unit, the average value of the supply-demand of the three types of CES in each township was calculated. We then classified the top 20% as hot spots by the quantile method; 20%–80% were classified as warm areas, and the last 20% were classified as cold areas (Meng et al., 2020; Shen et al., 2019). This method was used to map CES supply-CES demand in Suzhou, and then a map of CES supply-CES demand was compared to evaluate

Table 2
CES supply and demand assessment indicators.

Environmental variables	Data source	Calculations
Distance to the CES sites	POI point of CES sites (mywis.cn)	Euclidean distance to CES sites
Elevation	GDEM V3 30 M resolution digital elevation data (www.gscloud.cn)	Direct access
Distance to road	Road networks in Suzhou (www.openstreetmap.org)	Euclidean distance to roads
Distance to rivers and lakes	Rivers and lakes in Suzhou (www.openstreetmap.org)	Euclidean distance to lakes and rivers
Vegetation coverage index	Land use type (globeland30.org)	Calculate by ENVI
Landscape diversity index	Land use type (globeland30.org)	Calculate by Fragstats4.2.1
Habitat quality index	Land use type (globeland30.org)	Calculate by Invest3.9.2
Land use type	Land use type (globeland30.org)	Habitat quality
Population density	Population density (www.gisrs.cn)	Direct access
Night-time Satellite Data	Night-time Satellite Data (www.gisrs.cn)	Direct access

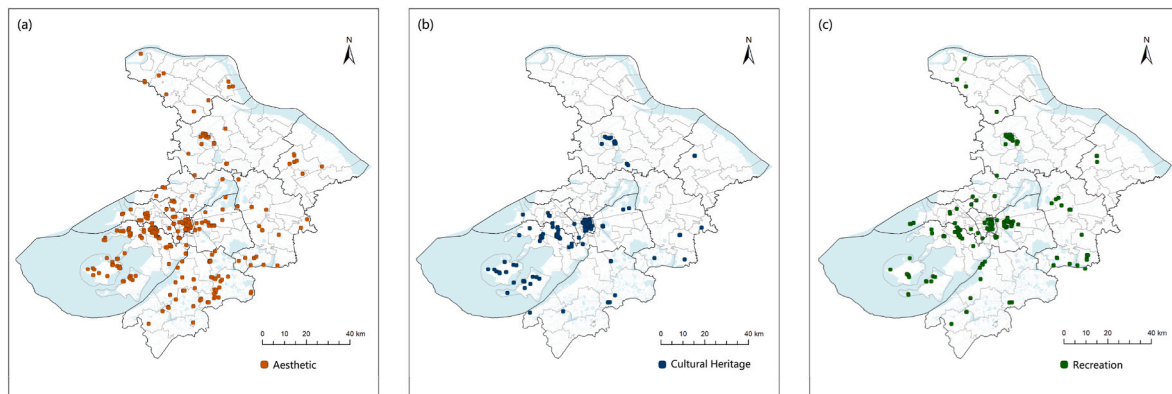


Fig. 4. The distribution of three types of photo spots in Suzhou (a. aesthetic CES; b. cultural heritage CES; c. recreation CES).

the balance situation of oversupply, overdemand, and supply-demand balance. Then, the proportion of the area between the three types of CES was calculated by township.

4. Results and analysis

4.1. Spatial patterns of CES supply and demand

The MaxEnt model predicts the potential CES supply more accurately. For prediction results, the AUC values of the three types of CES were as follows: aesthetic AUC = 0.873, cultural heritage AUC = 0.907, and recreation AUC = 0.871. It is generally believed that if AUC > 0.85, the model runs well and has high reliability. The AUC values for the results in this study were all above 0.85, which indicates high reliability. From the contribution (c) and importance (i) of the CES environmental variables, it can be seen that the key factors affecting the CES supply include elevation, distance to the CES site, and vegetation coverage index. However, variables such as distance to rivers and lakes and land use types contributed less (Table 3).

According to the map of CES supply, on the whole, the high-supply areas of CES were mainly distributed in Gusu District, around Taihu Lake Scenic Areas in the northeastern part of Wuzhong District, the southern part of Huqiu District, and the Yushan Scenic Area, which is in the middle of Changshu. Specifically, the distribution of high-supply areas for aesthetic and recreation services has a high degree of overlap; they are largely distributed in Gusu District, around the Taihu Lake Scenic Area in the northeastern part of Wuzhong District, and in the southern part of Huqiu District (Fig. 5). The location of high aesthetic supply areas was closely related to the distribution of mountainous areas in the region. For example, the overall CES supply capacity in Changshu was relatively low, but it formed a supply hot spot on Yushan Mountain, which was mostly due to the good landscape conditions provided by the mountainous scenic. At the top of the mountain, tourists could have a broader view. The relatively intact protection of mountain habitats can also provides a highly natural aesthetic value. Recreation services were closely related to the distribution of attractions such as parks and mountains, which provides a variety of recreational activities, such as mountain climbing and sightseeing. Chinese scholar gardens, parks and other cultural attractions are densely distributed in Gusu District, which contributes a basis for recreation service supply hotspots. The mountain scenic area formed a hotspot for the supply of recreation services at the Xishan Island Scenic Area in the middle of Wuzhong District. The shores of Taihu Lake and Xishan Island, where waterfront recreation and mountain outdoor sports are integrated, also exhibit a high recreation service supply capacity and have become focal points of tourism development in Suzhou. Cultural heritage service hotspots were mainly distributed in Gusu District and the Taihu Lake Scenic Area in the northeastern part of Wuzhong District, primarily related to the

distribution of historical and cultural heritage sites. Gusu District, as the main district of Suzhou city, is rich in history and culture and densely distributed with traditional scholar gardens, historical and cultural blocks, temples and other attractions. In recent years, as the Suzhou municipal government continues to strengthen efforts to implement historical and cultural heritage protection policies, a series of urban renewal projects have been carried out to renovate historical and cultural spaces with the characteristics of Suzhou Jiangnan water towns, which could further provide higher-quality cultural heritage services.

According to the spatial distribution of the CES demand map (Fig. 6), the high-demand areas of CES were mainly concentrated in Gusu District, around the Taihu Lake Scenic Area in the northeastern part of Wuzhong District, the southern part of Huqiu District, and the Yushan Scenic Area in the middle of Changshu. Gusu District, as we mentioned above, is the core destination for most tourists who visit Suzhou, featuring traditional scholar gardens and water towns, as well as scenic spot development, infrastructure improvements and other related aspects. Therefore, Gusu District shows a high level of demand in all three types of CES. The areas with high demand for aesthetic services were primarily concentrated in Gusu District and the northeast of Wuzhong District. The visual attractiveness of the local cultural landscape, such as traditional scholar gardens and water towns, also made Gusu District a popular destination for tourists to perceive aesthetic. Meanwhile, urbanization intensifies the people's longing for nature, the urban forest and wetland visual healing effect makes suburban mountain parks, and wetland parks become aesthetic service high-demand areas. The areas with a high demand for recreation services were largely located in Gusu District, around Taihu Lake Scenic Area in the northeastern part of Wuzhong District, Yushan Scenic Area in the middle of Changshu and the southeast of Kunshan. The demand for recreation services is closely related to the construction of recreational infrastructure in mountain parks. The addition of a mountain foot system provides hiking and cycling paths for outdoor enthusiasts. The areas with high demand for cultural heritage services were largely concentrated in Gusu District, around the Taihu Lake Scenic Area in the northeastern part of Wuzhong District and the Yushan Scenic Area in the middle of Changshu. People's demand for historical and cultural elements is mainly reflected in historical and cultural relics, traditional scholar gardens other cultural elements that create a perception of the historical and cultural atmosphere of the city.

4.2. Supply-demand balance analysis and optimization suggestions of CES in Suzhou

4.2.1. Supply-demand balance analysis

On the whole, the oversupply in Suzhou is the most evident in the cultural heritage services, and the noticeable demand shortage appears in both the aesthetic and recreation services (Fig. 7). The spatial distribution of the supply and demand imbalance for each type of CES is shown in Table 4.

4.2.2. CES supply-demand and tourism function mechanism elaboration

Compared to suburban counties, there are policy inclinations, more publicity, and better transportation and infrastructure in Gusu District tourist attractions. The CES supply and demand balance area in Gusu District comprises a relatively large area, while the imbalance areas are largely distributed in the suburban or county areas far from the city center. This is related to the highly concentrated distribution of scenic nodes within Gusu District. It presents the radioactive characteristics of Gusu ancient city as the core and continuously extends to the peripheral zone, forming a spatial structure pattern of "one core and multiple spots" (Fig. 8). At present, tourism development in the central urban areas of Suzhou tends to be saturated. With the increase in demand for tourism diversity and the expansion of information access channels, tourism in the suburbs and county areas around urban centers has gradually appreciated. However, the preference of tourist flow volume to the

Table 3
Environmental variable contributions.

Environmental variable	Aesthetic		Cultural heritage		Recreation	
	C	I	C	I	C	I
Distance to the aesthetic sites	30.9	32.8	/	/	/	/
Distance to the cultural heritage sites	/	/	/	/	16.4	14.2
Distance to the recreation sites	/	/	51.4	16.7	/	/
Elevation	39.2	9.9	14.3	14	36.4	3.1
Distance to road	0.7	5.8	3.7	25.3	5.3	17.7
Distance to rivers and lakes	1.0	1.9	0.4	0	12.3	12.7
Vegetation coverage index	3.1	9.9	1.3	2.5	13.6	38.9
Landscape diversity index	0.4	0.2	3.3	9.7	0.2	0.6
Habitat quality index	11.6	4.4	1.0	4.3	0.6	1.1
Land use type	1.8	2.4	21.1	21.8	11.1	8.3
Population density	5.9	14.2	2.6	4.2	2.8	0.5
Night-time Satellite Data	5.4	18.4	0.9	1.5	1.3	2.8

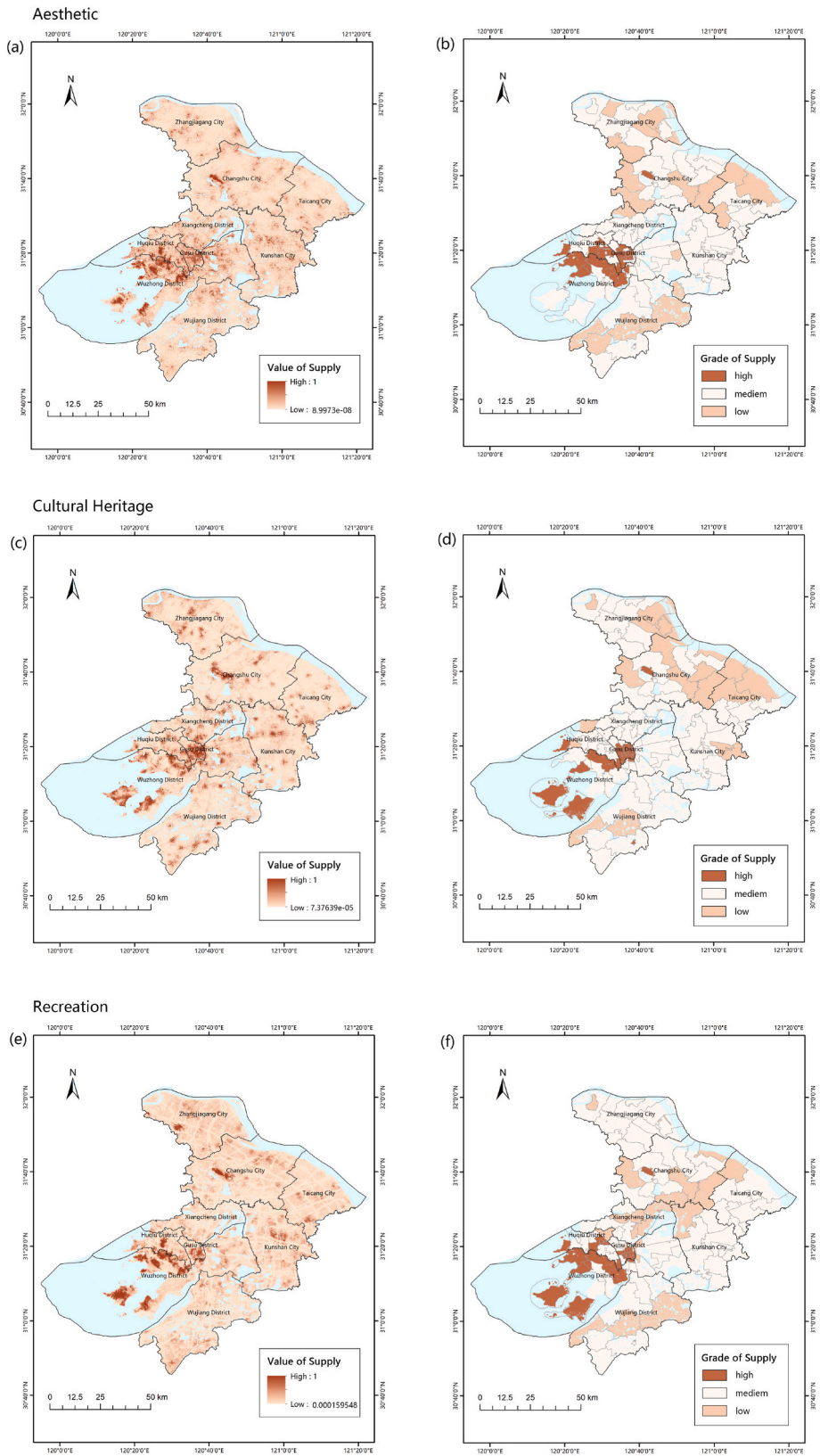


Fig. 5. Three types of CES supply in Suzhou (a, c, e are the distribution of each type of CES supply; b, d, f are the classification results of each type of CES supply based on the statistical average of the county).

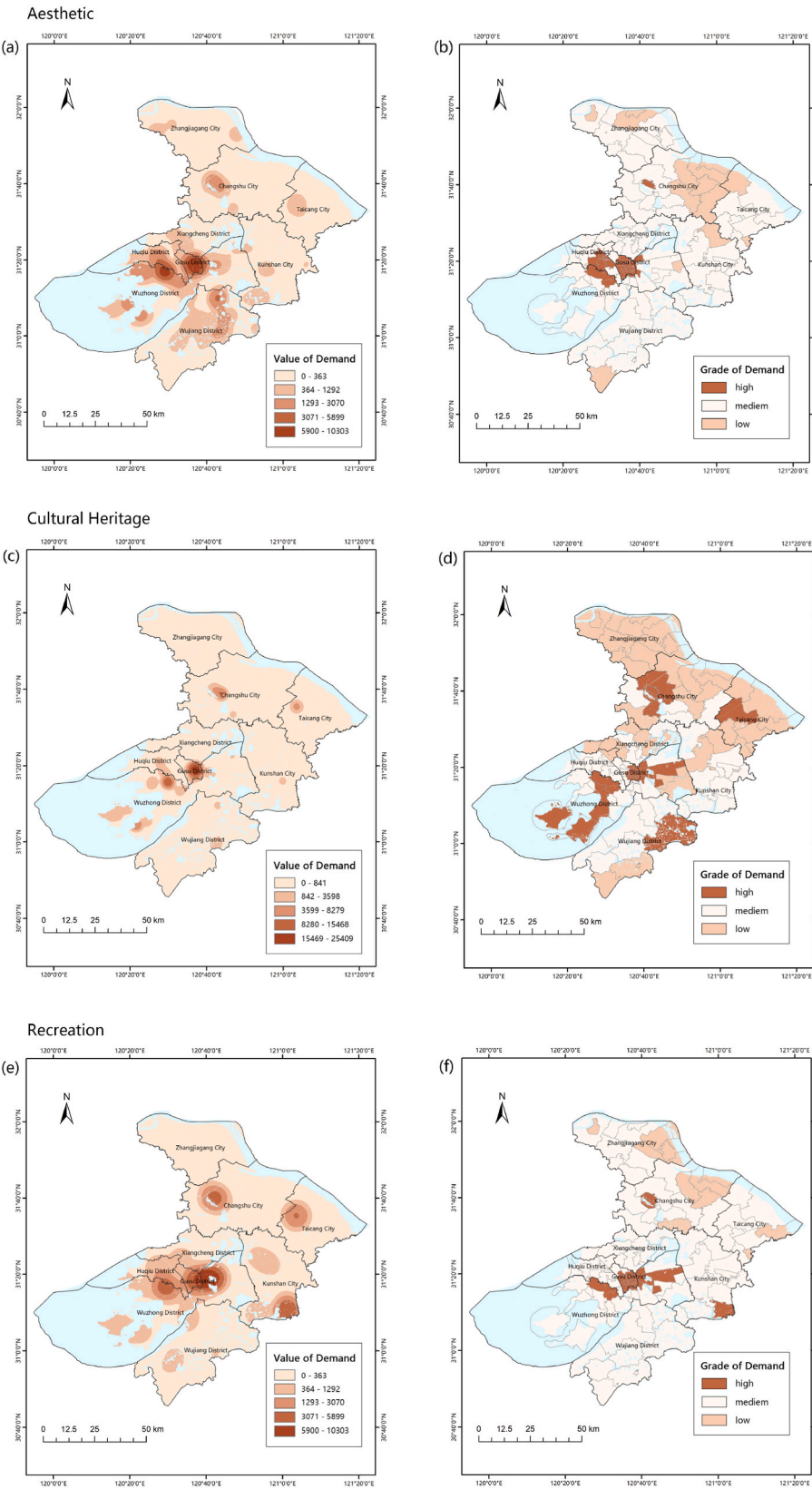


Fig. 6. Three types of CES demand in Suzhou (a, c, e are the core density distribution of each type of CES demand; b, d, f are the classification results of each type of CES demand based on the average value of county statistics).

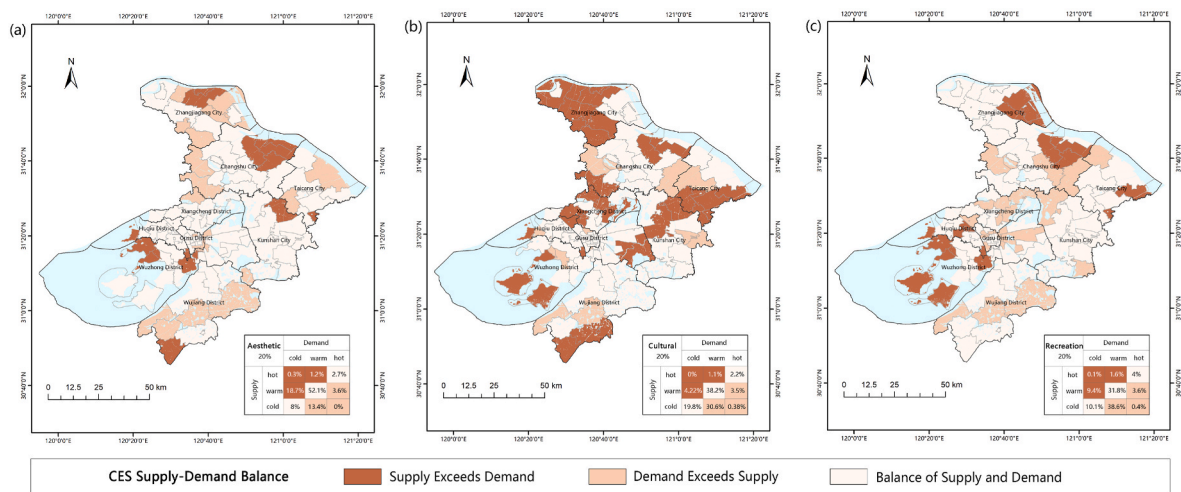


Fig. 7. Supply-demand balance and imbalance of three types of CES in Suzhou (a: aesthetic CES; b: cultural heritage CES; c: recreation CES).

Table 4
Three types of CES supply-demand relationships in Suzhou.

CES Type	Supply-demand imbalance type	Area proportion	Typical region
Aesthetic	Oversupply	16.9%	Northeast and north of Wuzhong District, south of Wujiang District, northeast of Changshu, north of Zhangjiagang, northeast of Kunshan
	Overdemand	21.5%	Central of Wujiang District, west of Changshu, east of Zhangjiagang
Cultural heritage	Oversupply	36.1%	Central of Wuzhong District, southern of Wujiang District, northeast of Changshu, west of Zhangjiagang, north of Kunshan, south of Taicang and the most of Xiangcheng District
	Overdemand	12.0%	Central of Wujiang District, west of Changshu, west of Taicang
Recreation	Oversupply	15.1%	Central of Wuzhong District, northeast of Changshu and part of northeast of Zhangjiagang
	Overdemand	21.9%	Central of Wujiang District, east of Changshu and most of Xiangcheng

suburbs and countryside could not bring corresponding benefits if these areas are improper in tourism protection and development. There are two scenarios for the impact of tourism protection and development on CES spatial imbalance (Fig. 9). Overdemand scenario: Some scenic spots adopt rapid development approaches to attract overflowing tourist flow volume from the urban center at the expense of the environment and the destruction of local cultural heritage. Overdevelopment puts tourism destinations in a state of CES overdemand, which will lead to the collapse of the entire tourism system once the threshold of environmental capacity is exceeded. Oversupply scenario: These areas are rich in natural landscape, cultural heritage and regional cultural features, but are difficult to attract tourists due to a lack of support for policy and investment. Meanwhile, a lack of exploration of tourism resources and the problem of insufficient service infrastructure make it difficult to meet the increasingly diversified demand of visitors, and inadequate development has left tourism destinations in these areas in CES oversupply (Jun & Xiaobin, 2022).

By analyzing the response mechanism of tourism protection and development to CES spatial imbalance, key scenic spots in the

imbalanced area can be identified. The area around the Taihu Lake Scenic Area of Wuzhong District presents a CES oversupply of cultural heritage and recreation, including the Dongshan and Xishan areas, where tourism and infrastructure are underdeveloped due to their location in the suburbs. However, its landscape type varies, and landscape quality is high, which provides high CES supply capacity; therefore, it has a high potential for tourism development. In central Wujiang District, an overdemand CES imbalance is evident. Ancient towns such as Tongli and Zhouzhuang are densely clustered in the region, and there is a serious situation of overdevelopment. Excessive commercialization and homogeneous competition made tourists unable to establish emotional links with ancient towns, and the demand for CES in ancient towns gradually decreased. As a result, the region presents an overdemand imbalance in all three CES (Fig. 7). Therefore, it is necessary to adopt a strategy of protective development which can reasonably balance the mismatch of the supply-demand of CES with the dynamic balance of supply-demand and sustainable tourism growth.

4.2.3. *Balanced supply-demand tourism planning strategy*

By analyzing the characteristics of scenic spots in three types of CES imbalance areas in Suzhou and combining the results of the supply-demand assessment and the characteristics of each area, we established a landscape spatial structure of six cores, three belts and three zones (Fig. 10). The six cores include the core of the comprehensive developed downtown, the core of Jiangnan watertown, the cultural core of the ancient town, the cultural tourism core of Taihu Lake, the core of Yangcheng Lake tourism, and the core of natural therapeutic areas in Changshu. The three belts include the belt of the Grand Canal cultural heritage, the belt of ancient Watertown cultural tourism, and the belt of dynamic leisure tourism along the Yangtze River. The three zones include the Taihu Lake ecological vacation and tourism zone, Lake Swing ecological leisure and tourism zone, and ancient town historical and cultural tourism zone.

5. *CES supply-demand balance at the scale of scenic spots*

Since the regional CES assessment cannot analyze the current tourism resource, transportation and environmental conditions of the sites in the county at a finer scale, this paper conducts a scenic spot scale analysis and proposes a site-specific supply-demand balance strategy.

5.1. *Supply-demand balance analysis of Xishan scenic spots*

Xishan Island belongs to the Taihu National Scenic Area. Compared with the core tourism area of Suzhou, development time on Xishan

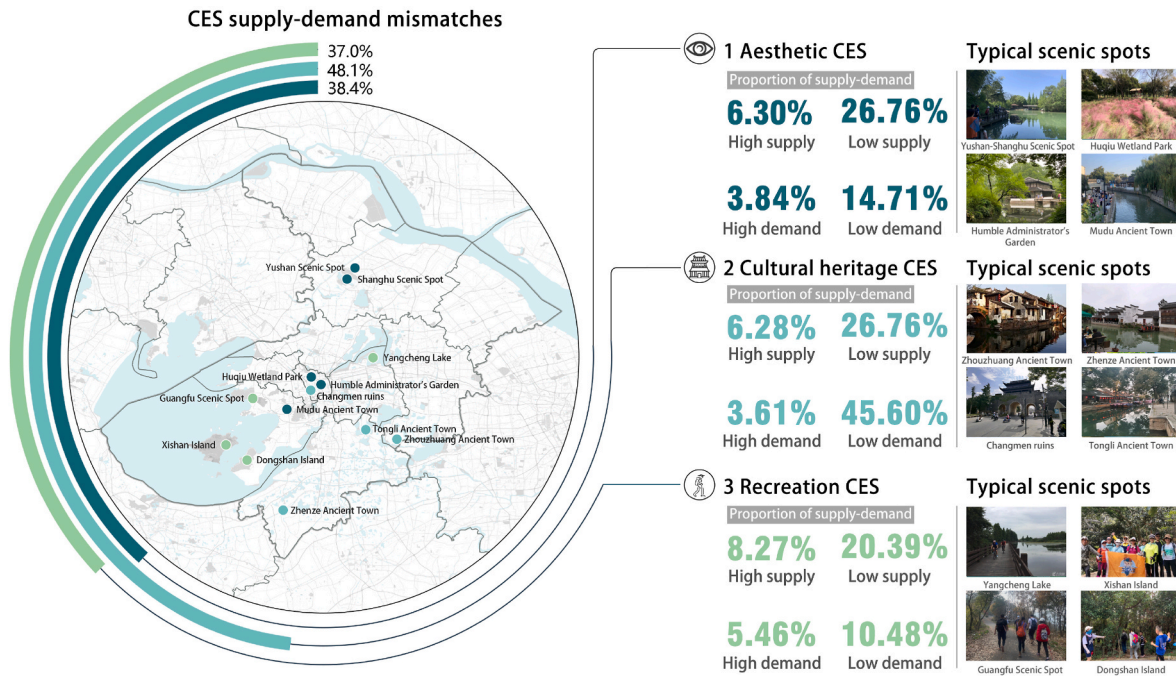


Fig. 8. Proportion of three types of CES in the area of supply-demand cold and hot spots and distribution of typical scenic spots in Suzhou.

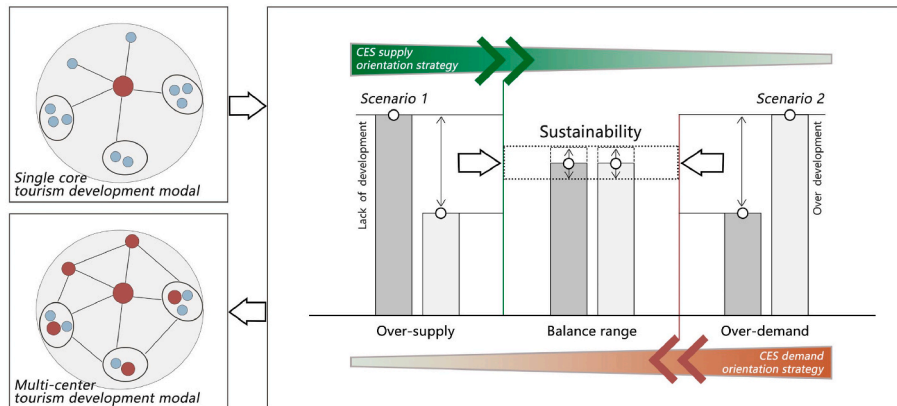


Fig. 9. The CES supply and demand interaction in tourism can be balanced and optimized through supply-oriented and demand-oriented strategies. This shift allows for a transition from a single core tourism development model to a more efficient multicenter model.

Island is short, and development level is low due to its relatively remote location. However, the ongoing regional integration development policy of Suzhou provides opportunities for its further development and protection. According to the three types of CES supply-demand maps for Xishan Island (Fig. 11), there were obvious supply-demand imbalances in the central, northeastern, southeastern and western parts of the island (Fig. 12). Among them, the aesthetic and recreation CES in the central part both indicated an oversupply in spatial distribution. The aesthetic and cultural heritage were overdemanding in the east, the cultural heritage was oversupplying in the southeast, and the cultural heritage and recreation were oversupplying in the west (Table 5).

In recent years, with the official announcement of the 'Suzhou Taihu Ecological Island Guidance', the tourism popularity of Xishan Island continues to increase, which further aggravates the supply-demand imbalance of CES on the island, and the contradiction between ecological protection and development has become increasingly prominent. Located in the middle, Piaomiao Peak is the highest point on Xishan Island, which ranks first among the seventy-two peaks in the Taihu Lake area. A healthy ecosystem results in a strong CES aesthetic and

recreation supply ability. However, due to the discontinuity of cultural memory, the lack of current development, and its low recognition in the tourism market, there is a lower CES demand for the public (Long, Zhenfang, & Dongye, 2020). In contrast, due to the high level of urban construction and economic development in the northeast area, traditional villages such as Houbu village have developed with high standards. Basic service facilities in the area are relatively complete, and the landscape vision is good, which makes it the core area for the aesthetic and cultural heritage CES demand. However, the CES supply in this area is noticeably insufficient. There are many mountains southeast of the island whose recreation supply capacity is high. The aggregation of Luohan Temple, Xiaoxia Bay, Dongcai and other traditional villages inside the mountain provides high cultural heritage CES. The CES of cultural heritage and recreation in the western part is characterized by oversupply. Historical and cultural relics such as traditional villages are distributed along the shore and close to the main road, which has a strong CES supply capacity. However, the demand for CES is low due to insufficient exploration of historical and cultural resources and the restriction of development intensity under protection measures.

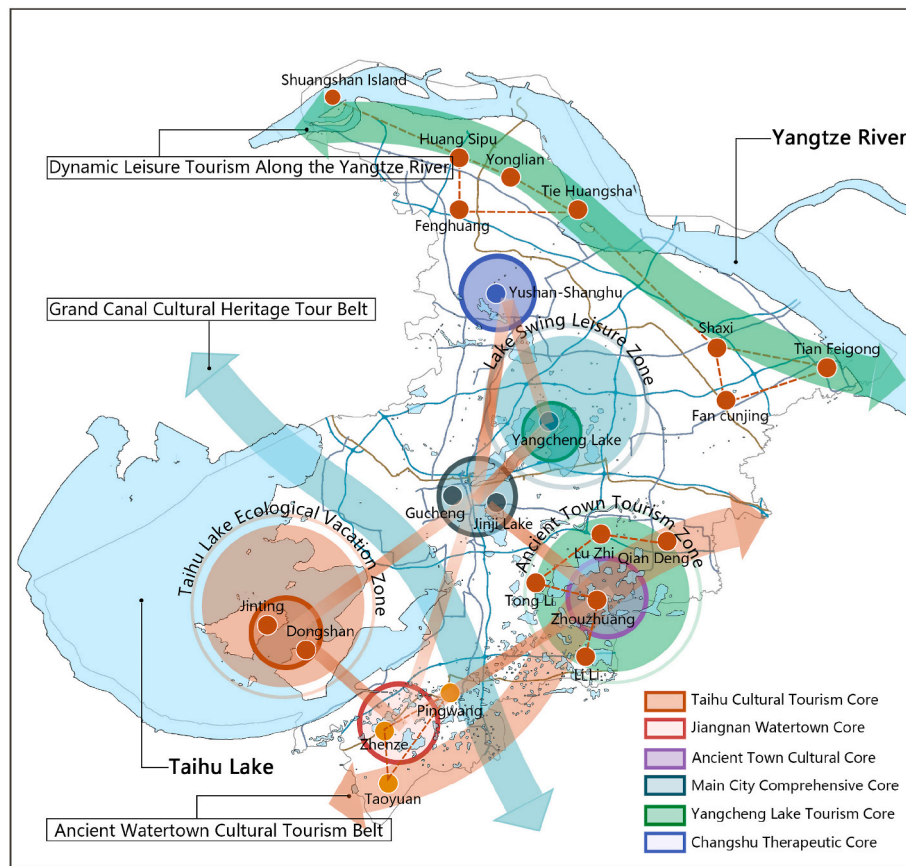


Fig. 10. Landscape spatial structure optimized for the balance supply-demand mismatches in Suzhou. A landscape spatial structure of six cores, three belts and three zones is constructed.

5.2. Supply-demand balance strategies for Xishan scenic spots

Due to the imbalance between the supply and demand of CES on Xishan Island, a spatial structure of the landscape with one belt, two axes and four zones is proposed (Fig. 13). Among them, the belt refers to traditional village tours, two axes include the axis of mountain tourism and interactive agriculture, and four zones include integrated forested mountain villages, mountain dock and lodge cultural, mountain fitness and agricultural tourism. The forested village integrated area should strengthen the development of Piaomiao Peak and its surrounding scenic areas, and establish connections with Han Village, Tangli Village and other villages. Connecting the historical and cultural relics to form the traditional village tour belts, Western traditional villages should strengthen the construction of infrastructure and combine with the linear distribution characteristics of CES supply and take full advantage of their lakefront location to incorporate waterfront leisure projects such as fishing and Fisherman's Family Tourism. The mountain fitness area should integrate with the mountain foot path system to create a Mantou mountain - Silong mountain - Luohan deck high-quality mountain-climbing route. High CES demand attracts agricultural sightseeing areas, such as Taihu flower fields and happy farms, to help protect the natural environment and perform protective development for surrounding farmland and wetlands, such as supporting fruit picking, science education, and other interpretative agricultural activities, to promote CES supply stability. At the same time, the monitoring of ecological indicators should be strengthened to avoid negative impacts on the environment by unsuitable development.

6. Discussion

6.1. The importance of the landscape in connecting CES and tourism

The spatial mapping of CES reflects the priority of regional management (Plieninger et al., 2015; Zoderer et al., 2016), which is conducive to the formulation of land use and planning management policies in the future (Plieninger et al., 2015) and is of great significance for the sustainable development of tourism (Bachi et al., 2020). Here, the landscape is regarded as the medium of tourism resources and the means to attract tourists (Terkenli, Skowronek, & Georgoula, 2021). Combining CES assessment with landscape management and planning and optimizing landscape structure is an important way to identify priority areas for tourism; meanwhile, it could promote the development and protection of tourism destinations (Plieninger et al., 2015). Based on the driving factors of CES supply-demand and the resistance factors of tourism development, landscape planners can put forward corresponding comprehensive planning and management strategies for the CES supply-demand imbalance areas (Albert et al., 2016) to counterpoise the CES supply-demand mismatch between tourism destinations and their surroundings.

Social media photos contain rich geographic information (Heipke, 2010). Flickr and other social media platforms have recently been widely used for CES assessment and mapping, which reflect the public's perceived preference for nature and can be acquired quickly and in large quantities. These advantages provide the possibility for efficient evaluation of CES (Zhang, Huang, Zhang, & Buhalis, 2020). However, the accessibility and user attributes of social media largely determine the scope of research objects, which may lead to a lack of representativeness of the samples (Keeler et al., 2015; Tenerelli et al., 2016), which could bring some limitations to the study (Zhang et al., 2020). Maxent can be

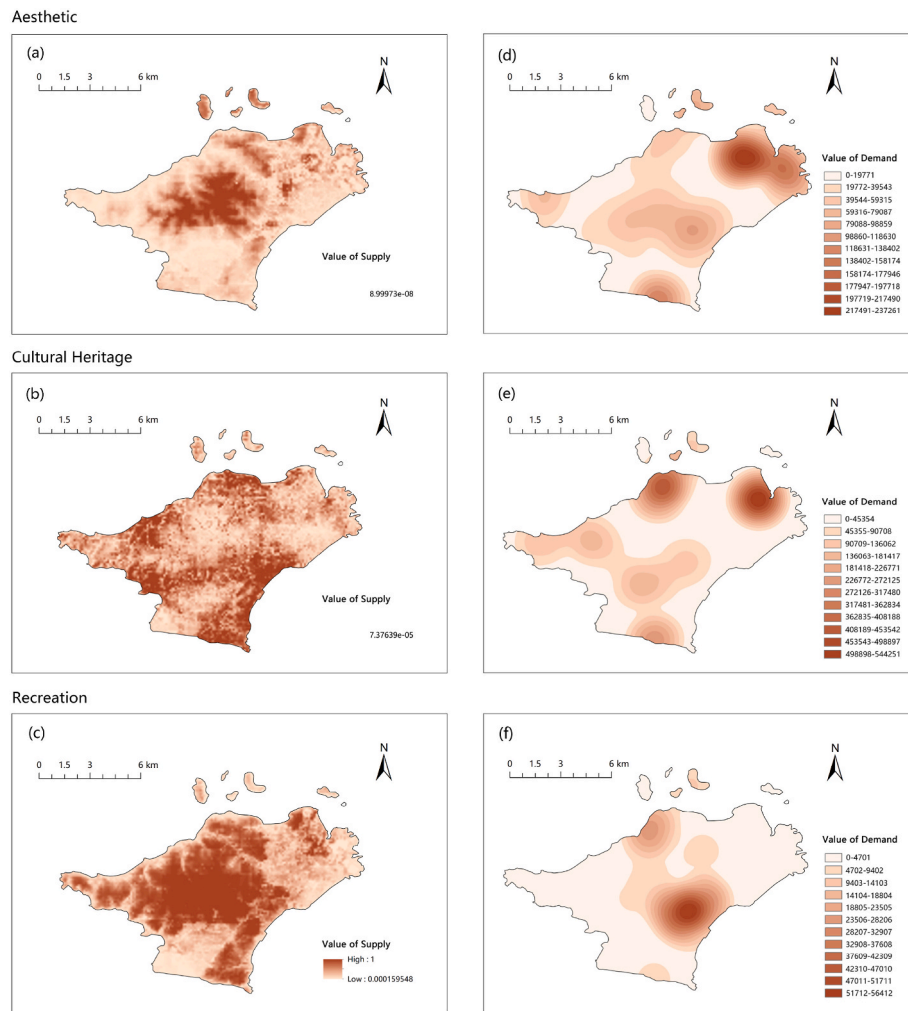


Fig. 11. Spatial pattern of CES supply and demand in Xishan Island scenic spots (a: CES aesthetic supply; b: CES aesthetic demand; c: CES cultural heritage supply; d: CES cultural heritage demand; e: CES recreation supply; f: CES recreation demand).

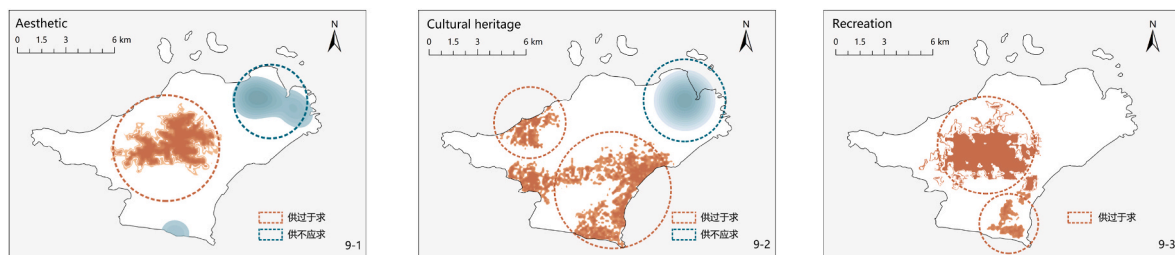


Fig. 12. The supply-demand balance and imbalance of three types of CES in Xishan Island scenic spots (a: aesthetic CES; b: cultural heritage CES; c: recreation CES).

employed to model a potential distribution based on existing data. This method allows for a comparatively objective CES spatial distribution map. Correlating user preferences with environmental factors overcomes the difficulty of CES mapping due to the inclusion of subjective and intangible assessments (Phillips, Anderson, & Schapire, 2006; Arslan & Öricü, 2021.). In this study, geotagged social media photos were used to map the supply-demand distribution of three types of CES at multiple scales, providing unique guidance for the formulation of tourism protection and development policies.

6.2. The relationship between CES assessment and mapping, landscape management and planning, and tourism protection and development

The spatial relationship between the supply and demand of ecosystem services plays a large role in landscape management and planning (Baró et al., 2017). CES assessment and mapping localize the evaluation results to real geographic spaces through spatial modeling (Dramstad, Tveit, Fjellstad, & Fry, 2006; Havinga, Bogaart, Hein, & Tuia, 2020; Langemeyer, Calcagni, & Baró, 2018). Counties with an obvious imbalance between supply and demand determine areas where supply and demand should be coordinated (Shi, Shi, Zhou & Fang, 2020; Meng et al., 2020). Then, the analysis is conducted on a more refined scale to extract the focused scenic spots with an imbalance between

Table 5

The Distribution of hotspots on Xishan Island and the relationship between the supply and demand of the three types of CES.

Supply-demand hotspots	Core scenic spots	Supply and demand situation		
		Aesthetic	Cultural heritage	Recreation
Midlands	Piaomiao Peak, Xishan Geological Museum	Oversupply	Balance	Oversupply
Northeast	Houbu Village, Farmville	Overdemand	Overdemand	Balance
Southeast	Linting Plum Blossom Park, Mingyue Bay, Yuwang Temple, Carved Flower Building	Balance	Oversupply	Oversupply
West	Han Village, Tangli Village	Balance	Oversupply	Oversupply

supply and demand in the county, identify the key landscape elements in the focused scenic spots and optimize those landscape structures (Schirpke et al., 2021). Through cooperation between regional management officers and tourism destination planners, key points of strategy formulation at different scales could be determined (Santarém, Saarinen, & Brito, 2020) to put forward strategies to guide tourism protection and development accordingly. For tourism destinations with oversupply, development opportunities should be seized to determine their own characteristics (Martinez-Harms et al., 2018), and their tailored tourism attractions should be enhanced by optimizing the construction of landscape features and strengthening publicity efforts (Gliozzo, Pettorelli, & Haklay, 2016; Ungaro, Häfner, Zasada, & Piore, 2016). For tourism destinations with overdemand, we should pay attention to monitoring indicators of habitat quality to measure the level of environmental carrying capacity, avoiding problems such as habitat destruction caused by excessive demand (Meng et al., 2020).

Meanwhile, the connectivity of surrounding high-supply areas should be strengthened to ease demand pressure (Bing et al., 2021; Chung, Pan, Zou, & Liu, 2018).

On one hand, previous research on the evaluation of supply and demand in CES has predominantly relied on a single scale, resulting in assessment outcomes that offer limited guidance for regional tourism resource management and scenic area planning and design practices. On the other hand, landscape management and planning have not been adequately integrated into CES supply and demand evaluations, making it difficult to specify spatial allocation strategies for CES in tourism destinations. To address these issues, this paper proposes a multi-scale tourism development guidance system that encompasses “CES evaluation and mapping – landscape management and planning – tourism conservation and development” (Fig. 14). Through multi-scale CES supply and demand assessments, key imbalances between supply and demand in specific areas can be identified, enabling the formulation of management and planning strategies for CES supply and demand allocation that are tailored to local conditions. This framework not only enhances the theoretical research on spatial allocation of CES supply and demand but also provides a practical approach for spatial allocation at multiple scales within the Suzhou, while offering a replicable model for integrated tourism regions and sustainable tourism development globally.

6.3. Limitations and future research

The framework proposed in this paper provides a new direction for spatial optimization and tourism planning at the urban-scenic spot scale in Suzhou. However, since the main users of the ‘six foot’ outdoor platform used in this study are outdoor enthusiasts, the crowd characteristics caused a lack of data sources in some remote areas. It is also difficult to focus on the fine scale of a specific tourist attraction, such as ancient towns, villages, and parks, if the data are collected from social media (Richards & Friess, 2015; Cheng, Van Damme, Li, & Uyttenhove, 2022). Meanwhile, it is challenging to explore the effects of local policies, cultural background, socioeconomic conditions and other

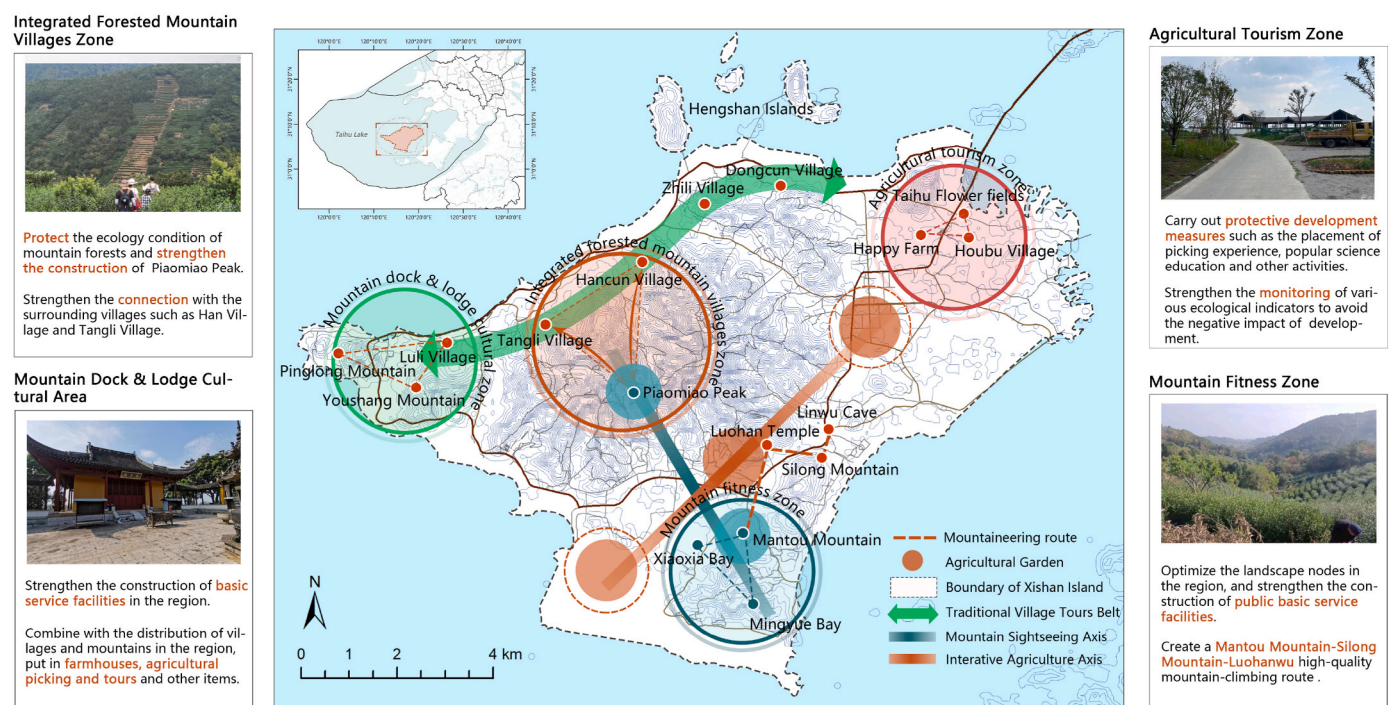


Fig. 13. Landscape spatial structure optimized for the balance supply-demand mismatches in Xishan Island scenic spots. A spatial structure of the landscape with one belt, two axes and four zones is proposed.

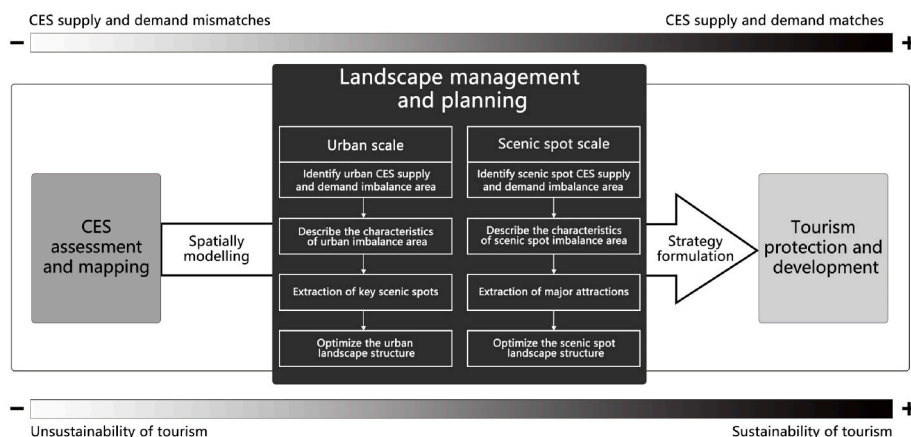


Fig. 14. A system for CES assessment and mapping – landscape management and planning – tourism protection and development'. Among them, landscape management and planning provide key information at the urban scale and scenic spot scale, which is a spatial organization approach to connect CES and tourism. This would increase CES supply and demand matches and the sustainability of tourism.

socioeconomic factors on CES assessment and mapping to develop detailed design strategies (Bullock, Joyce, & Collier, 2018; Felipe-Lucia, Comín, & Escalera-Reyes, 2015; Zhou et al., 2022). Thus, combining social media data with other CES mapping methods to form a more complete co-action planning and design system might be a new direction for future exploration. For example, integrated social media and participatory mapping methods (Depietri, Ghermandi, Campisi-Pinto, & Orenstein, 2021; Moreno-Llorca et al., 2020) to assess CES might provide a new way to design research for different categories of tourism destinations at a finer site scale. Moreover, in future studies, research on social media photos could be further extended in combination with photo comments, semantic analysis and other methods to extract landscape elements that affect tourists' perceptual preferences. This study was conducted based on a single group of visitors. In the future, multi-group studies (Yuqing, Weibin, Xueer, Dongjin & Hui, 2022; Darvill & Lindo, 2016) could be conducted to determine the differences in the perceived needs of different groups for CES and balance the needs of multiple groups in the process of planning and design strategy formulation. It is of great significance for tourism destinations to alleviate the contradiction between the demands of multigroup CES and the need to realize their harmonious coexistence.

7. Conclusion

Although some scholars have studied the spatial distribution of CES at the urban scale in the past, few studies have focused on the supply-demand situation and have carried out assessments and analyses at the scale of scenic spots. This study analyzed the spatial distribution characteristics of different types of CES supply-demand and identified the balance situation between supply-demand among tourists. The results showed that there was a clear imbalance between supply and demand in Suzhou. We then formulate landscape management and planning strategies for CES areas with oversupply and undersupply, which are important media connecting CES and tourism protection and development. Through the identification of key landscape elements in CES imbalance areas, based on the balancing of CES mismatches, sustainable tourism protection and development could be achieved by landscape management and planning. Based on geotagged social media photos, by employing MaxEnt and ArcGIS to map the CES supply and demand at multiple scales for aesthetic, cultural heritage, and recreation, we proposed a system of 'CES assessment mapping – landscape management and planning –tourism protection development'. This system provides a theoretical and practical basis for the multiscale application of CES assessment in landscape management and planning, which provides a new way for the protection and development of

tourism destinations. In future research, comprehensive evaluation and decision-making of CES can be achieved by supplementing diverse metadata, such as Points of Interest (POI), comments, and data obtained through questionnaires and interviews, while taking into account the preferences of different population groups. This implication is not only useful for this study region but also of great significance for other urban regions which aim to achieve sustainable tourism.

CRedit authorship contribution statement

Chang Li: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. **Qifan Liang:** Software, Data curation, Formal analysis, Visualization, Writing – original draft. **Beiduo Lin:** Visualization, Formal analysis, Investigation. **Jun Zhai:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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